

Q1. (a) Identify the error and briefly explain how to correct the error in each of the following program segments:

(i) `int x = 7, y = 8;`
 `int *xptr = &x, result;`
 `int *yptr = &y;`
 `result = *xptr * yptr;` (4 marks)

(ii) `float m = 1.0;`
 `float *mptr = &m;`
 `float **mpptr = &mptr;`
 `//use mpptr to access m value and display`
 `cout << "m value is:" << *mpptr;` (4 marks)

(iii) `char ary[5] = {'a', 'b', 'c', 'd', 'e'} ;`
 `char *ptr;`
 `ptr = ary;`
 `for(i=0; i<5; i++){`
 `ary = ary + i;`
 `cout << *ary << "\t";`
 `}` (4 marks)

(b) Given the following declarations:

```
int x[10] = {11, 12, 13, 14, 15, 16, 17, 18, 19, 20};  
int *ptr1, *ptr2;  
ptr1 = &x[5];  
ptr2 = ptr1 - 2;
```

What will be printed out in the following statements?

(i) `cout << ptr1[-2];` (4 marks)

(ii) `cout << ptr2[-1] * ptr1[+3];` (4 marks)

Q2. (a) (i) Answer the following questions based on the linked list in Figure 1.0 below.

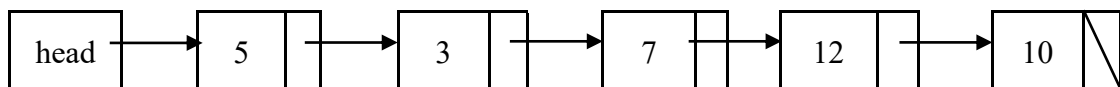


Figure 1.0 Linked List

- (i) Draw a diagram of the resulting linked list if the following lines of code are executed on the linked list in Figure 1.0 above. (5 marks)

```
Node* newNode = new Node(9)

newNode->next = head->next->next;
head->next->next = newNode;
```

- (ii) Draw a diagram of the resulting linked list if the following lines of code are executed on the linked list in Figure 1.0. (5 marks)

```
Node *temp = head;
while (temp->next != NULL) {
    temp->item = temp->next->item;
    temp = temp->next;
}
temp->next = head;
```

- (b) Based on the following segment of code below, state what is the output of the following cout statement or whether the cout will cause error. You need to justify your answer if it cause error. (10 marks)

<pre> struct personal { char phoneNumber[11]; char address[50]; char city[15]; char state[3]; char zipCode[6]; }; struct customer { char name[30]; int customerNumber; personal p; //name, customerNumber, phoneNumber, address, city, state, zipcode customer(char*, int, char*, char*, char*, char*, char*); }; </pre>	<pre> void main() { customer cust = { "John Lim", 3298, "0123672018", "No. 23, Jalan Timah", "Kampar", "Perak", "31900" }; customer *ptr; ptr = &cust; cout << "\n\nptr->name[3]: " << ptr->name[3]; cout<<"\n\ncust.p.address[8]:" << cust.p.address[8]; cout << "\n\n(*ptr).p.zipCode:"<< (*ptr).p.zipCode; cout << "\n\ncust.p.customerNumber:" << cust.p.customerNumber; cout << "\n\nptr->p->state:" << ptr->p->state; } </pre>
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cout statement	Output or Error
cout << "\n\nptr->name[3]:" << ptr->name[3];	
cout << "\n\ncust.p.address[8]:" << cust.p.address[8];	
cout << "\n\n(*ptr).p.zipCode:" << (*ptr).p.zipCode;	
cout << "\n\ncust.p.customerNumber:" << cust.p.customerNumber;	
cout << "\n\nptr->p->state:" << ptr->p->state;	

- Q3. (a) What is the output of the following program? The display function will display the content of the stack in this format [1 2 3] (where 1 is the first node and 3 is the last node in the list). (9 marks)

<pre>int main() { float x = 0.0; float y = 0.0; Stack stack1; //Store float stack1.push(3.2); stack1.push(2.0); cout << "\n\n*****A*****\n"; stack1.display(); stack1.pop(x); stack1.pop(y); stack1.push(y - x); stack1.push(7.9); stack1.push(5.7); cout << "\n\n***B***\n"; stack1.display(); //continue in second column</pre>	<pre>//continue from first column stack1.pop(x); stack1.pop(y); stack1.push(y + x); cout << "\n***C***\n"; stack1.display(); stack1.pop(x); stack1.pop(y); y *= x; cout << "\n**D***\n"; stack1.display(); cout << "\n\nx:" << x << "y:" << y << "\n"; return 0; }</pre>
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(b) What is the output of the following program segment?

(11 marks)

```
Queue queue;
string temp;
string fruit[6] = { "Apple", "Orange", "Durian", "Banana", "Coconut", "Mango" };

for (int i = 0; i < 6; i++)
    queue.enqueue(fruit[i]);

while (queue.size() > 1) {
    for (int i = 1; i <= 2; i++) {
        queue.dequeue(temp);
        if (i < 2)
            queue.enqueue(temp);
        else
            cout << temp << " is out.\n";
    }
}
queue.dequeue(temp);
cout << temp << " is the last fruit in the queue.\n";
```

- Q4. (a) Write a recursive function void Minimum(int arr[], int n, int &min) to find minimum value in an array that store positive integer. The n is the size of array and min will store the minimum value in the array. Based on the execution of the statements below, min will have value 1 after function call. (8 marks)

```
int arr[10] = { 3,98,2,9,10,56,32,9,1,23 };
int min = arr[0];
Minimum(arr, 10, min);
cout << "\nMinimum value: " << min << "\n"; //print 1
```

- (b) A Lucas function to generate a series of integer with 2, 1, 3, 4, 7, ... is given below. Indicate each of the section below whether it is a base case or recursive segment. (6 marks)

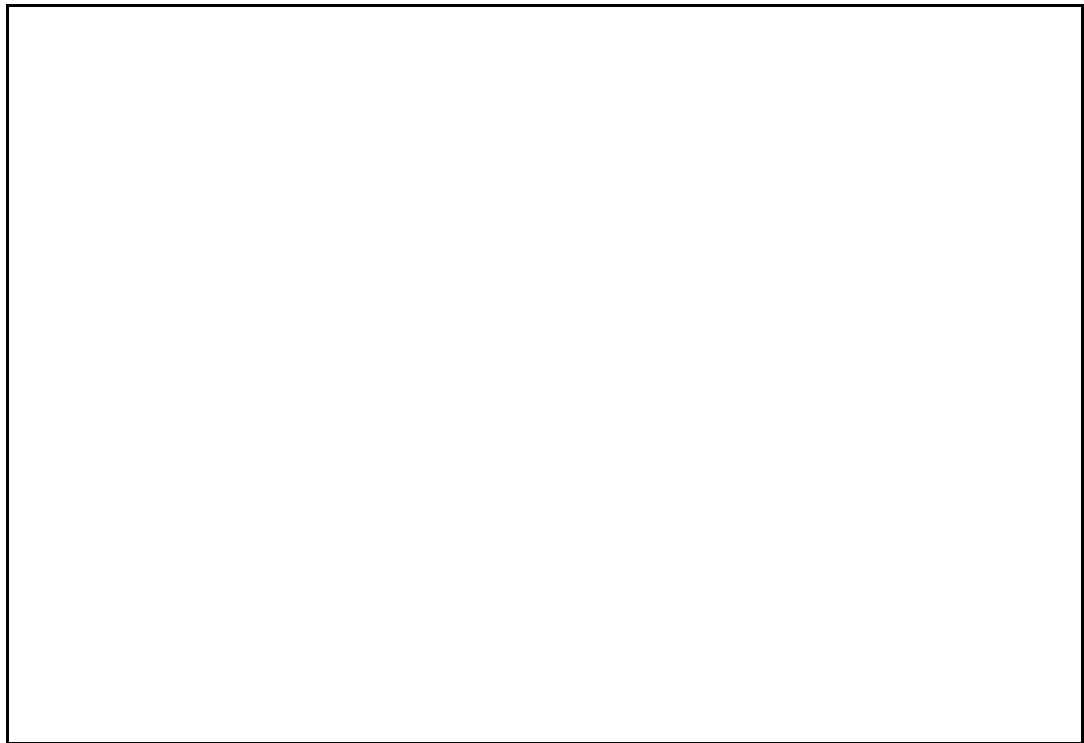
```
int Lucas (int n) {  
    if (n == 0)           //Section A  
        return 2;  
    else if (n == 1)      //Section B  
        return 1;  
    else                  //Section C  
        return Lucas(n - 1) + Lucas(n - 2);  
}
```

- (c) The function to find a greatest common divisor (gcd) is given below. For example, gcd for 10 and 25 will be 5. (6 marks)

```
int gcd(int a, int b) {  
    if (b == 0)  
        return a;  
    if (a == 0)  
        return b;  
    return (gcd(b, a % b));  
}
```

What is the value of result and how many times the gcd function will be called (including first function call) if gcd is call below:

```
return = gcd(3, 26));
```



- Q5. (a) The following are the values to be inserted to an empty Binary Search Tree.

58, 89, 38, 51, 17, 8, 92, 87, 93, 24, 75

- (i) Draw the Binary Search Tree based on the order of the value inserted above. (4 marks)



- (ii) List the order of the nodes visited using Preorder traversal manner. (2 marks)



- (iii) Write a function `bool BT::insert(type father, type son)` to insert a node with known father. You may write your function with 1 normal case and 2 special cases to deal with out-of-memory and empty tree. (7 marks)



- (iv) What will be printed if the Binary Search Tree drawn in Q5 (a) (i) above (before deleting) is used to call the function guessOutput below? Assume that cur is pointing to root and value is 39 when the function is called. (7 marks)

```
void BST::guessOutput(BTNode *cur, int value) {  
    if (cur == NULL)  
        return;  
    if (cur->item > value) {  
        guessOutput(cur->left, value);  
        cout << cur->item << " ";  
    }  
    guessOutput(cur->right, value);  
}
```

